# Effect of Temperature on Photovoltaic Cell Efficiency

Laboratory Report · Adam Nowak, Kasia Wiśniewska, Michał Kowalczyk · v14

## Abstract

This report examines how cell temperature affects the conversion efficiency of a monocrystalline silicon cell. Measurements were taken between 25 °C and 65 °C under constant illumination of 1000 W/m².

## 1. Objective

Determine the temperature coefficient of efficiency for the test cell and compare it with values reported in the literature.

## 2. Theoretical Background

Photovoltaic efficiency decreases as temperature increases. [1] The open-circuit voltage falls by roughly 2.2 mV per kelvin, while the short-circuit current rises only slightly [2]. As a result, the fill factor and the maximum power point both shift downward.

- Voc decreases by about 2.2 mV/K (lecture_05.pdf, p. 22)
- Isc increases by about 0.05 %/K
- Net efficiency loss of about 0.41 %/K

## 3. Methodology

We put the cell on the plate and made it hotter step by step and wrote down the numbers.

## 4. Results

Table 1 — Electrical parameters of the test cell at 1000 W/m²

| T (°C) | Voc (V) | Isc (mA) | η (%) |
|---|---|---|---|
| 25 | 0.612 | 34.8 | 19.4 |
| 35 | 0.589 | 35.1 | 18.6 |
| 45 | 0.566 | 35.4 | 17.8 |
| 55 | 0.543 | 35.8 | 17.0 |
| 65 | 0.520 | 36.1 | 16.2 |

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Improve writing | Shorten | Expand | Explain | Find evidence | Check claim | Add citation

We put the cell on the plate and made it hotter step by step and wrote down the numbers.

**Suggestion · Improve writing** uses only your text

~~We put the cell on the plate and made it hotter step by step and wrote down the numbers.~~
The cell was mounted on a thermostatic plate and heated in 5 °C steps; voltage and current were recorded at each step.

Accept | Reject | Edit

## 4. Results

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| T (°C) | Voc (V) | Isc (mA) | η (%) |
|---|---|---|---|

CONTENT ORIGIN
Human written
AI generated / rewritten
Source-derived claim
Analysis-derived claim

## Improve writing

SELECTED TEXT
“We put the cell on the plate and made it hotter step by step…”
Human written | 1 sentence

**What changed, and why**

**Passive, past tense**
Lab methods are written in passive voice.

**Specific equipment**
“the plate” → “a thermostatic plate”.

**Step size added**
5 °C steps — instructions.pdf, p. 5.

**What was measured**
Names voltage and current instead of “the numbers”.

**Nothing new was added**
The rewrite uses only your sentence and the 5 °C step size from instructions.pdf, p. 5. No new claims.

Try another version
More formal | Shorter
Keep my wording | Try again

**You stay in control.** Suggestions never replace your text on their own. Accept, reject or edit — every accepted change is recorded in the paragraph's provenance.

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**AI draft — not in your document yet** Grounded in 3 sources

The efficiency of a silicon cell depends on the junction temperature. As temperature rises, the intrinsic carrier concentration grows, which lowers the open-circuit voltage by about 2.2 mV per kelvin [2].

The short-circuit current increases only slightly because the band gap narrows. The net effect is a loss of roughly 0.41 % of efficiency per kelvin [1], as the laboratory instructions also state [3].

Insert draft | Insert and edit | Regenerate | Discard

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CONTENT ORIGIN
Human written
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Source-derived claim
Analysis-derived claim

## Generate section

Section: 2. Theoretical Background

Sources to use
- lecture_03.pdf — PDF · 31 pp.
- instructions.pdf — PDF · 9 pp.
- smith_2025.pdf — PDF · 24 pp.
- lee_2024.pdf — PDF · 18 pp.
- Internet — off

Length: ~700 words (~150 – ~1500)

Citations: Required | Where possible | Off

Tone: Academic

Generate draft

**Draft ready — 698 words**
6 citations · every sentence traced to a page. The draft is shown as a suggestion; your document is unchanged.

**Not in a source?** Sentences without support are marked “Needs citation” instead of being invented. Internet search is off for this draft.

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- Isc increases by about 0.05 %/K
- Net efficiency loss of about 0.41 %/K

**1 Smith et al. 2025**
smith_2025.pdf · page 14

*"...conversion efficiency fell by 0.41 % per kelvin over the 25–75 °C range..."*

Open source | View context | Replace

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SELECTED PARAGRAPH
"Photovoltaic efficiency decreases as temperature increases. The open-circuit voltage falls by roughly 2.2 mV per kelvin..."

Content origin
**Human written**
Adam Nowak · Sep 28

AI modifications
**Improved writing — Sep 30**
Accepted by Adam · 1 sentence

Sources
1 **smith_2025.pdf**
Smith et al. (2025) · page 14
2 **lecture_05.pdf**
Lecture 5 · page 22

Analysis
None — no analysis result is used here.

History of this paragraph
- **Citation [2] added**
  Michał · Oct 1
- **Improved writing accepted**
  Adam · Sep 30
- **Written**
  Adam · Sep 28

CONTENT ORIGIN
- Human written
- AI generated / rewritten
- Source-derived claim
- Analysis-derived claim

Paragraph history | Check claim